Huawei Access Optical Transport Failure Resolution SOP for YEL-ACC-17 at Yelahanka Sub-station

Node: YEL-ACC-17 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6933

Description

This SOP addresses the isolation and resolution of an optical transport failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) occurring on the Huawei Access equipment (Node ID: YEL-ACC-17, located at Yelahanka Sub-station). The failure is characterized by the ALARM_6933 alarm code with a Minor severity level. This failure domain is specific to the Access role within the Huawei equipment, where the Optical Transport Network (OTN) protocol governs the data transmission between nodes. The underlying protocol mechanics involve the Optical Channel Transport Units (OCH-TUs), which carry data over the fiber optic cables. The SFP/XFP modules, responsible for transmitting and receiving optical signals, are prone to degradation due to environmental factors, hardware malfunctions, or fiber cuts. The OTN protocol's robust error detection and correction mechanisms trigger the ALARM_6933 alarm when the received optical signal power falls below a predetermined threshold.

Pre-requisite Checks

- Verify that the Yelahanka Sub-station is accessible and dispatch remote hands if necessary
- Confirm that the redundant link status is operational
- Ensure that the Huawei equipment is running the latest software version
- Check for any recent hardware or software changes that may have contributed to the failure

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to high temperature or humidity	High
Fiber cut or damage	Medium
Rx/Tx power loss due to misaligned fiber connections	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical link brief, Expected Output: Optical link brief information, Description: This command displays a brief summary of the optical link status, including the received and transmitted optical power levels, which can help identify if the issue is related to the SFP/XFP module or fiber connectivity.

Command: display optical sfp, Expected Output: SFP/XFP information, Description: This command displays detailed information about the SFP/XFP module, including its temperature, voltage, and optical signal quality, which can indicate if the module is functioning correctly.
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Command: display optical fiber, Expected Output: Fiber information, Description: This command displays information about the fiber optic cables, including their length, type, and status, which can help identify if the issue is related to fiber cuts or damage.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical sfp , Description: This command replaces the SFP/XFP module with a new one, which can resolve issues related to SFP/XFP degradation.

Step 2: Step: Realign the fiber connections, Command: realign optical fiber , Description: This command realigns the fiber connections to ensure proper alignment and secure connections, which can resolve issues related to Rx/Tx power loss due to misaligned fiber connections.

Step 3: Step: Repair or replace the fiber optic cables, Command: repair optical fiber , Description: This command repairs or replaces the fiber optic cables, which can resolve issues related to fiber cuts or damage.

Post-Check Verification

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Command: display optical link brief, Expected Output: Optical link brief information, Description: This command verifies that the optical link status has been restored and the received and transmitted optical power levels are within the normal range.

Command: display optical sfp, Expected Output: SFP/XFP information, Description: This command verifies that the SFP/XFP module is functioning correctly and the temperature, voltage, and optical signal quality are within the normal range.

Command: display optical fiber, Expected Output: Fiber information, Description: This command verifies that the fiber optic cables are functioning correctly and the length, type, and status are within the normal range.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: undo replace optical sfp , Description: This command restores the original SFP/XFP module if the replacement does not resolve the issue.

- Step: Realign the fiber connections to the original position, Command: undo realign optical fiber , Description: This command realigns the fiber connections to the original position if the realignment does not resolve the issue.

- Step: Replace the fiber optic cables with the original ones, Command: undo repair optical fiber , Description: This command replaces the fiber optic cables with the original ones if the repair or replacement does not resolve the issue.